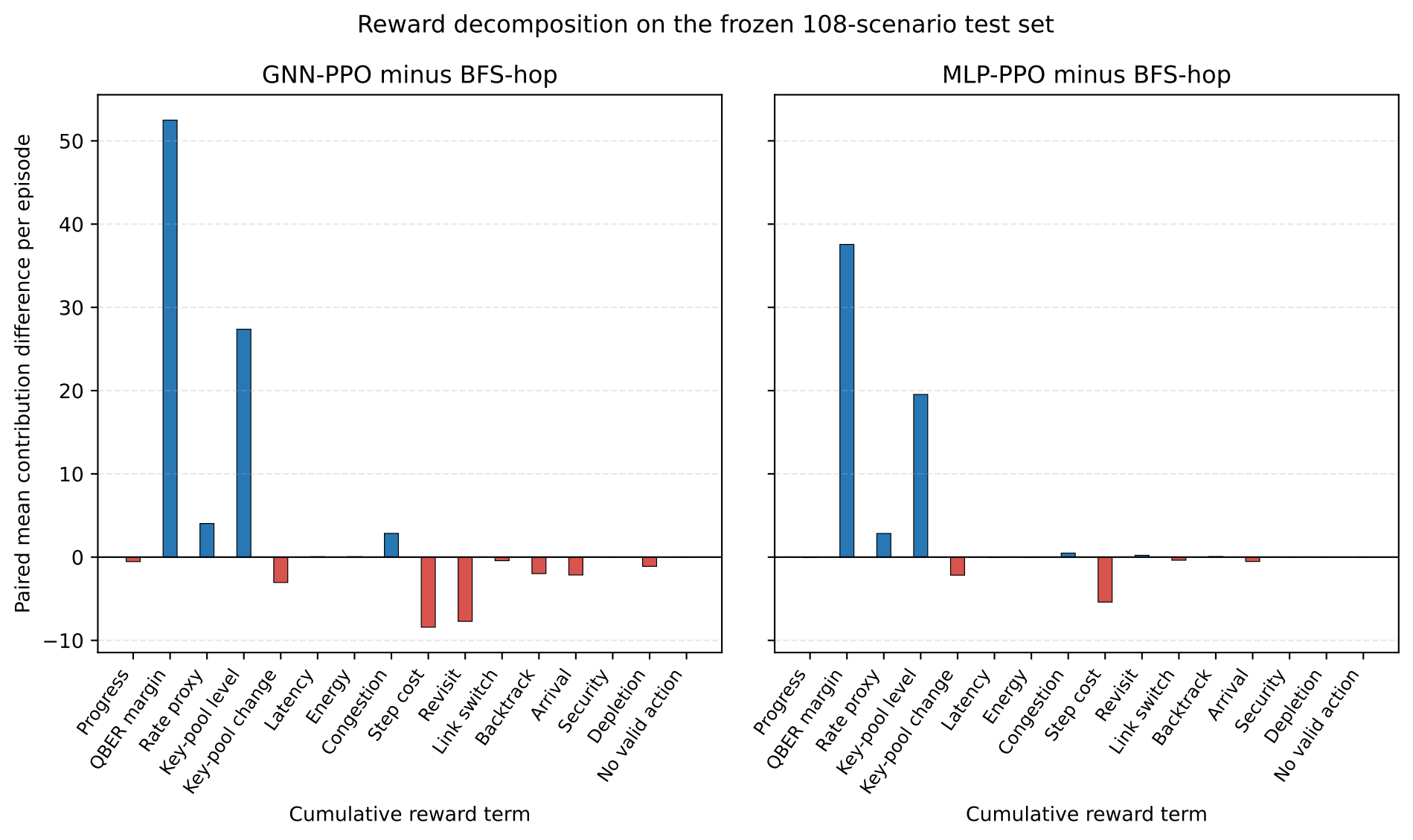
Reward decomposition on the frozen 108-scenario test set
GNN-PPO minus BFS-hop
MLP-PPO minus BFS-hop
Paired mean contribution difference per episode
50
40
30
20
10
0
−10
Progress
QBER margin
Rate proxy
Key-pool level
Key-pool change
Latency
Energy
Congestion
Step cost
Revisit
Link switch
Backtrack
Arrival
Security
Depletion
No valid action
Cumulative reward term
Cumulative reward term